

Week 45

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```
try:
    import fysisk_biokemi
    print("Already installed")
except ImportError:
    %pip install -q "fysisk_biokemi[pyodide] @ https://au-mbg.github.io/fysisk-biokemi/
    wheels/ fysisk_biokemi-0.2.2-py3-none-any.whl
```

1 Python: Calculations

```
import numpy as np
```

(a) Simple calculations

Evaluate the following expressions using Python

- 10.5×4
- $5 + 10 + 15 + 12$
- $70 - 30 + 2$
- $420/10$

```
... # Replace ... with your code
...
...
...
```



Tip

Recall that the basic arithmetic operations are, in Python, represented as follows

- $+$ $\Rightarrow$ `+`
- $-$ $\Rightarrow$ `-`
- $/$ $\Rightarrow$ `/`
- $\times$ $\Rightarrow$ `*`
- x^y $\Rightarrow$ `x**y`

(b) Array calculations

The ability to perform operations on all elements (or data points) is incredibly useful when doing data analysis, for example when converting units. This is enabled by array calculations.

The cell below defines an array a with all the numbers from 0 to 9.

For each entry a_i in the array calculate the expression $b_i = a_i/3 + a_i^2$ and assign that to a new array b . Use the fact that you can do elementwise operations with an array - do not calculate it separately for every element of the array.

```
a = np.arange(0, 10)
```

```
b = ...
print(b)
```



Tip

With an array all the basic arithmetic operations can be done elementwise, so the expression $x^2 + 1$ can be done for all elements in an array as such

```
x = np.arange(0, 10)
y = x**2 + 1
```

Just like if x was just a single number. Similarly, expressions can involve adding arrays together like

```
y = x**2 + x
```

(c) Variables

Variables are an essential part of a program. Calculate the following where each intermediate is assigned to a variable

$$a = 5 \times 3$$

$$b = a + 9$$

$$c = \frac{4}{3}a + b - 2$$

```
a = ...
b = ...
...
print(a, b, c)
```

Bonus question: Why does c print as a decimal number when the others print without decimals?

(d) Parentheses

Parentheses are important, a misplaced or lacking parentheses can drastically change a calculation.

Using the variables a, b and c you calculate in the previous exercise, evaluate the following expressions

- $a \times (b + c)$
- $a \times b + c$
- $\frac{a+b+c}{b+c}$
- $(b - a) \left(\frac{1}{c} + 1 \right)$
- $(b - a) \left(\frac{1}{c+1} + 1 \right)$

```
q1 = ...
q2 = ...
q3 = ...
q4 = ...
q5 = ...
print(q1, q2, q3, q4, q5)
```

Tip

If you're in doubt having an extra set of unnecessary parentheses is better than missing a necessary set of parentheses. Just be careful to place them correctly.

(f) Functions

For repeated computations it's good practice to define a function.

Define a function that calculates the following expression

$$f(x) = \frac{Ax + B + C}{A + B}$$

Where $A = 10$, $B = 5$, $C = 2.5$

```
def fun_function(x):
    result = ...
    return result
```

Now evaluate the function for $x = 62.5$, $x = 629.25$, $x = 42$ and $x = 2025$

```
fun_output_1 = fun_function(...) # Your code replaces ...
fun_output_2 = ... # Same here
```

```
fun_output_3 = ... # And here
fun_output_4 = ... # And finally here.

print(f"{fun_output_1:.1f}")
print(f"{fun_output_2:.1f}")
print(f"{fun_output_3:.1f}")
print(f"{fun_output_4:.1f}")
```

💡 Tip

The input of a function will often be the independent variable and the functions calculates the independent variable.

Later when we get to regression/fitting this will be the case.

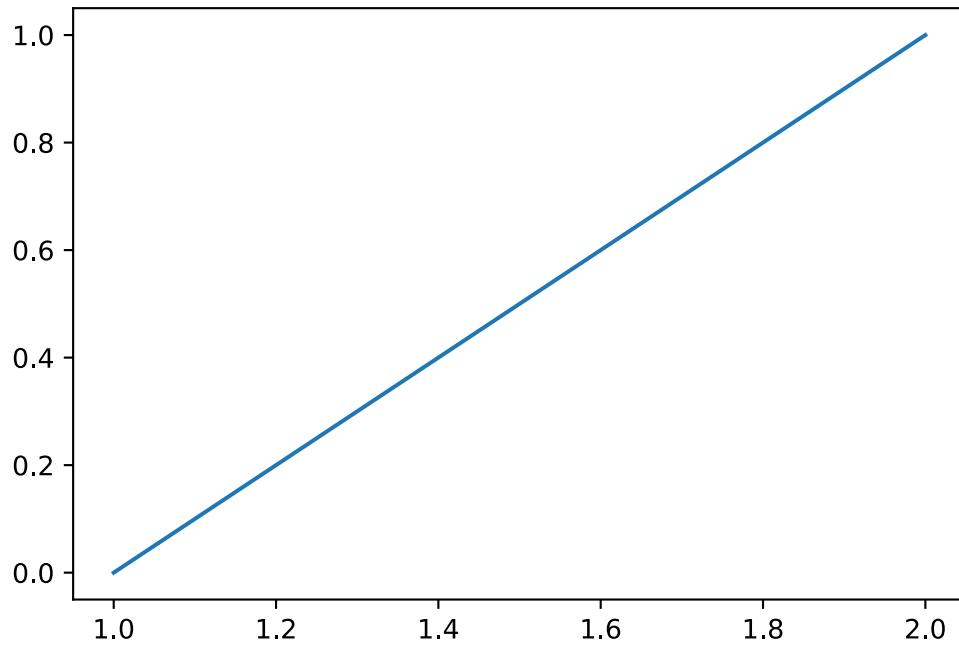
2 Python: Introduction to plotting

```
import numpy as np
import matplotlib.pyplot as plt
```

There are many ways to plot, we will be using the matplotlib library - but the concepts are similar across most ways of plotting.

We can make a plot of a straight line between two points (x_1, y_1) and (x_2, y_2) like so;

```
x = np.array([1, 2]) # This array contains all the x-coordinates
y = np.array([0, 1]) # And this contains all the y-coordinates.
fig, ax = plt.subplots(figsize=(6, 4))
ax.plot(x, y)
```

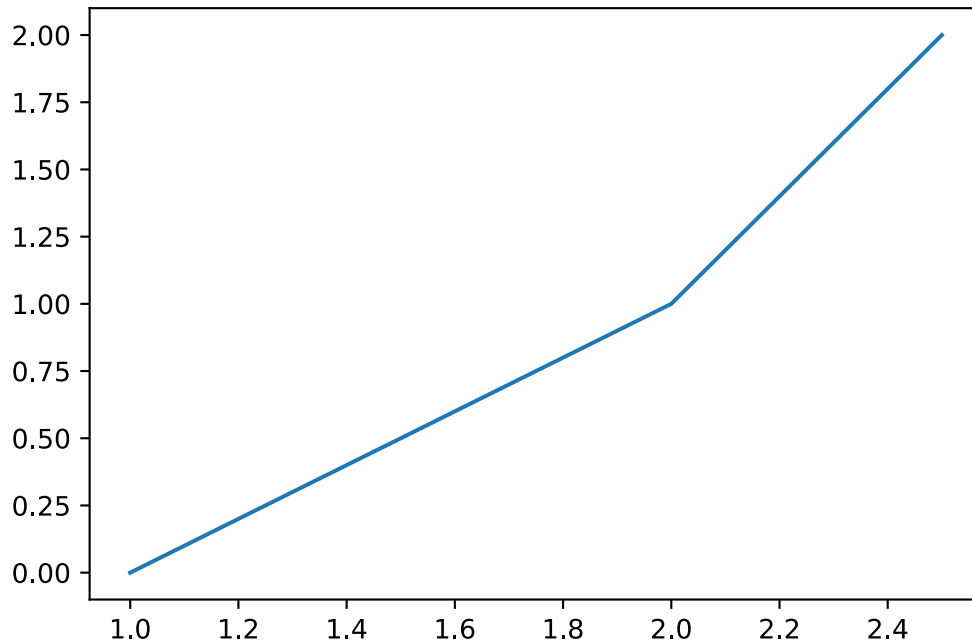


i Note

ax is a special type (like int, float, str), you can think of it as the box that contains the plot.

If we want to connect to a third point (x_3, y_3) we would instead write

```
x = np.array([1, 2, 2.5])
y = np.array([0, 1, 2])
fig, ax = plt.subplots(figsize=(6, 4))
ax.plot(x, y)
```



Notice that with three sets of points we plot **two** line segments.

(a) Trying to plot a square

Make a plot with lines between the following 4

- $P_1 = (0, 0)$
- $P_2 = (1, 0)$
- $P_3 = (1, 1)$
- $P_4 = (0, 1)$

Start by defining an array for both x and y

```
x = np.array([...]) # Replace ... with your code
# Do the equivalent for y
```

Now use `ax.plot` to make the plot

```
fig, ax = plt.subplots(figsize=(5, 5))
... # Replace with your code.
ax.axis('equal')
ax.set_xlim([-0.5, 1.5]) # Tells matplotlib to show us the region between -0.5 and 1.5 on
the x-axis.
ax.set_ylim([-0.5, 1.5])
```

Consider the following:

- Why is it not a square?
- How many points did you use?
- How many line segments were plotted?

(b) Actually plotting a square

In order to actually plot a square you will need to update x and y , they both need to contain five numbers such that four line segments are plotted.

```
x = np.array([...]) # Replace ... with your code
# Do the equivalent for y
```

Copy your code for making the plot from the previous exercise to the cell below

```
... # Put your copied code here.
```

(c) Plotting a function

Often the data we plot can broadly be thought of as originating from a function, that is we have

$$y = f(x)$$

Where f is the function, this might for example be an experiment that produces some output y given some input x or it might be a traditional mathematical function like $y = x^2$. Plotting this type of data is exactly the same as plotting a square - it just usually consists of many more data points resulting in many line segments producing a smooth looking curve.

Make a plot of the function

$$y = e^x$$

Note

The exponential function can be calculated using `np.exp` - likewise with the logarithm `np.log`.

```
x = np.linspace(0, 5, 50) # Make 50 points uniformly between 0 and 5.
y = ... # Your code here.

fig, ax = plt.subplots(figsize=(6, 4))
... # Your code to plot here
```

Once you've made the plot try changing the number of points in x by changing the last argument in the call to `np.linspace`.

- What happens if you have very few points?

Tip

Remember that `np.exp` is used to calculate the exponential function.

(d) Plotting multiple functions

Often we want to plot multiple functions in the same figure as it enables us to compare them. Luckily, this is quite simple!

Plot the function

$$y = e^{ax}$$

With $a = \left[1, \frac{1}{2}, 1.2\right]$ in the same plot producing three curves.

```
x = np.linspace(0, 5, 50) # Make 50 points uniformly between 0 and 5.
y1 = ...
y2 = ...
y3 = ...

fig, ax = plt.subplots(figsize=(6, 4))
ax.plot(x, y1)
ax.plot(x, y2) # To plot a second curve we just use ax.plot again
... # Plot the the third curve.
```

(e) Customizing plots

We will generally need to customize plots a little bit, for example the plot from the previous exercise doesn't have a label for the axis, there's no way of telling which curve is which and it has no title.

To add labels to the x and y axis we use the two functions

- `ax.set_xlabel('String with the name of the x-axis')`
- `ax.set_ylabel('String with the name of the y-axis')`

Adding information about the curves we plot can be done by also giving them a label, that is done as an argument to the `ax.plot` call, like so

```
fig, ax = plt.subplots()

ax.plot(x, y, label='Information about the curve.')
ax.legend()
ax.set_title('Title of plot')
```

Line 3

The `label` argument is given when the curve is plotted.

Line 4

`ax.legend()` is called to tell matplotlib to show the a 'legend' containing the label of all the plots that have been given one.

Line 5

Sets the title of the plot

Copy your code from the previous exercise and add labels for both the x and y axis aswell as each curve, for example you can label them according to the value of the parameter a like $a = 1$ etc.

```
... # Put some code here and try customizing it.
```

Ellipsis

i Note

There are many other customization options that can be added in the same way as the `label`, that is

```
ax.plot(x, y, option_name=option_value)
```

Some useful examples are

- `linestyle`: Controls if the plot is made using a full, dashed or dotted line using `-`, `--` and `:` respectively. So `ax.plot(..., ..., linestyle = ":")` produces a dotted line.
- `color`: Controls the color of the line - valid options are listed here https://matplotlib.org/stable/gallery/color/named_colors.html
- `linewidth`: Sets the width of the line - can be any number.
- `alpha`: Controls the transparency of the plot 0 being fully transparent and 1 being fully visible.

You can try some of these options for the plot you've made above if you want to. There are *many* other ways of customizing plots for different situations. The matplotlib gallery shows a number of them.

(f) Other types of plots: Scatter

Sometimes we don't want to connect each point with a line segment but just show the points in a *scatter* plot.

The next cell makes some data

```
x = np.random.uniform(low=0, high=10, size=100)
y = 2*x + np.random.normal(scale=2, size=100)
```

Try plotting it like for the previous exercises

```
# Put your code to quickly plot in this cell - no customization needed!
```

You will see a very strange plot like when you scribble out a word on a piece of paper. Clearly a line plot is not a very good way of showing this data! There are, at least, two ways of making a scatter plot

- `ax.plot(x, y, 'o')`: Quick and dirty way of just showing the points and not the lines.
- `ax.scatter(x, y)`: Function specifically for making these types of plots - with its own set of customization arguments.

Try either, or both, of these ways in the cell below

```
fig, ax = plt.subplots()
```

```
# Your code here.
```

3 Average properties of amino acids and proteins.

```
import pandas as pd
import numpy as np
from IPython.display import display
```

In the accompanying Excel file (averag-prope-amino-acids.xlsx), you will find a table that contains the molecular weight of the 20 common amino acid residues, i.e. their weight as residues in a peptide chain. Additionally, you will find their relative frequency in E. coli proteins, where a frequency of 0.01 means that this residue constitutes 1 % of the residues in a protein.

(a) Load the data file.

Load the averag-prope-amino-acids.xlsx dataset

```
from IPython.display import display
```

Use the cell below to load and display the dataset.

```
data_file_path = ... # Write the path to the dataset file
df = pd.read_excel(data_file_path)
display(df)
```

(b) Average molecular weight

Calculate the average molecular weight of a residue in a protein. To do this our procedure will be as follows

1. First calculate the residue weight times its frequency in proteins for each residue.
2. Sum these together to get the average molecular weight.

In the cell below finish the calculation of `weight_times_freq` by extracting the "MW of AA residue"-column and "Frequency in proteins" and multiplying them together.

💡 Tip

You can index in the dataframe by using the column name, for example to get the "MW of AA residue"-column you would do

```
col = df["MW of AA residue"]
```

Arrays also allows us to operate on every element in the array at the same time, so arrays can be added, subtracted, multiplied, divided, etc, see for example this figure

`data = np.array([1,2])`

data
1
2

`ones = np.ones(2)`

ones
1
1

`data`

1
2

-

`ones`

1
1

=

0
1

`data`

1
2

*

`data`

1
2

=

1
4

`data`

1
2

/

`data`

1
2

=

1
1

```
weight_times_freq = ...
```

💡 Tip

You can use `np.sum` to sum all values in an array. For example

```
array = np.array([1, 2, 3, 4, 5])  
sum_of_array = np.sum(array) # Gives 15
```

```
average_mw = ...  
print(f"{average_mw = :.3f}")
```

i Note

The syntax `f"{average_mw = :.3f}"` is just a way of printing the value with a nicer format, in this case we print the value to 3 decimal places. In Python these are called f-strings, you don't need to understand the details at the moment.

(c) Weight of 300-residue protein

What would the molecular weight of a 300-residue protein most likely be, if you did not know its sequence?

```
mw_300 = ...  
print(f"{mw_300 = :.3f}")
```

In many projects, you will be working with a mixture of proteins. This could for example be a cell lysate or a biological fluid for protein abundance analysis, or the early stages of a protein purification process. In these situations, you cannot work with a molecule specific extinction coefficient. Instead, we would use the average values, which we will determine below.

(d) Average concentration of residues

In many cases, we work with mixtures of proteins that do not have a defined molar extinction coefficient. Instead, we work with an average 'mass extinction coefficient', which we will deduce here.

Calculate the average concentration of amino acid residues in a protein mixture at 1 mg/mL.

```
c_residue_avg = ...  
print(f"{c_residue_avg = :3.3f} M")
```

(e) Absorbance of an average mixture of residues

Calculate the absorbance from such a mixture under the assumption that only Trp and Tyr contribute.

```
freq = df.set_index("Name")["Frequency in proteins"]  
f_trp = freq["Tryptophan (Trp/W)"]  
f_tyr = freq["Tyrosine (Tyr/Y)"]  
c_trp = ... # Erstat med din kode.  
c_tyr = ... # Erstat med din kode.  
  
print(f"{c_trp = :3.5f}")  
print(f"{c_tyr = :3.5f}")
```

```
A280_1mg_pr_ml = ... # Erstat med din kode.  
print(f"{A280_1mg_pr_ml = :3.3f}")
```

In the compendium, we mentioned a "rule of thumb" that 1 mg/mL protein at a path length of 1 cm has an A280 = 1. How well does this compare to your calculated value?

(f) Absorbance → concentration

For a cell lysate, you measure an absorbance of 0.78 at a path length of 0.5 cm. What is the protein concentration?

```
# Set known values:
A = ... # Unitless
l = ... # cm

# Calculate concentration:
conc_mg_pr_mL = ...
print(f"Protein concentration = {conc_mg_pr_mL:.3f} mg/mL")
```

4 UV spectrum of apo- and holo-myoglobin

```
import matplotlib.pyplot as plt
import pandas as pd
from IPython.display import display
```

In this exercise we will analyze the spectra of apo- and holo-myoglobin. The dataset is given in `uv-spec-apo-holo-myo.csv`.

(a) Load the dataset

Load the dataset with pandas. Set `data_file_path` to the path of the dataset file named above.

```
data_file_path = ... # Write the path to the dataset file
df = pd.read_csv(data_file_path)
display(df)
```

(b) Plot the spectra



Tip

To plot we use the `matplotlib` package. Plots are generally just straight lines connecting points, with enough points we get a smooth looking figure.

For example, to plot a line connecting three datapoints

```
fig, ax = plt.subplots()

x = np.array([1, 2, 3])
y = np.array([1, 4, 9])
ax.plot(x, y)
plt.show()
```

Line 1

This creates a `fig` and an `ax`, the `ax`-object is the box where our plot is created.

Line 3

Define the data to put on the x-axis.

Line 5

Does the actual plotting

Line 6

Shows the figure

There are *many* ways of customizing plots, you will see different ones in the exercises, but by no means all of them - if you are interested you can find more information on the `matplotlib` documentation.

You don't have to worry about the NaN values in the dataset when plotting, `matplotlib` just skips plotting that line segment.

Using `matplotlib` plot each spectrum in the same figure as line plots.

```
fig, ax = plt.subplots()

# This selects the 'holo_absorbance'-column and plots it
ax.plot(df['wavelength'], df['holo_absorbance'], label='Holo')

# Copy the line above and edit it to also plot the Apo absorbance
... # Your code here

ax.set_ylabel('Absorbance (AU)')
ax.set_xlabel('Wavelength (nm)')
ax.legend()
```

(c) Apo & Holo

Based on the spectra explain what ApoMb and HoloMb represent?

You have learned that pure proteins without any UV/Vis-absorbing prosthetic groups bound, have basically no absorbance at wavelengths above $\lambda > 320$ nm. Nevertheless ApoMb still show some absorbance above 320 nm. In this case, it can be explained by the fact that ApoMb was generated from HoloMb by a procedure that will not be explained here.

(d) Absorption above 320 nm.

Explain why ApoMb in the spectrum above absorbs light at $\lambda > 320$ nm.

(e) Efficiency estimate.

Give a rough estimate of the efficiency of the chosen procedure of heme-group removal. You may wish to look at the numerical values in the DataFrame.

(f) Isobestic points

Are there any isobestic points between the two spectra?

To get a better understanding of the causes for the different spectra, you can compare to literature. The figure below shows the absorbance spectra of three states of myoglobin.

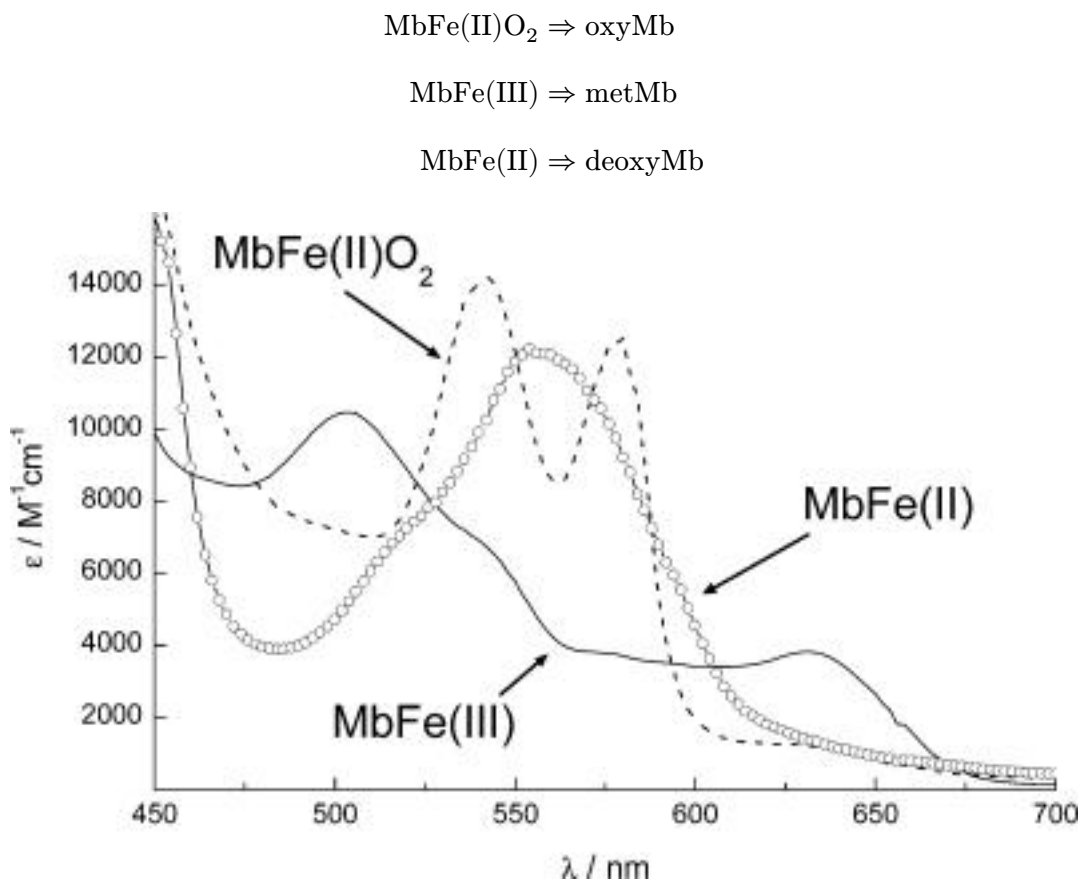


Figure 1: Absorbance spectra of different states of myoglobin. MbFe(II)O₂ is “oxyMb”, MbFe(III) is “metMb” and MbFe(II) is “deoxyMb”. Taken from Barham et al (2010); ChemRev; 110 (4); pp2313-2365

(g) Aged myoglobin

metMb (MbFe(III)) is normally described as ‘aged’ myoglobin. What does this mean in terms of the bound iron?

(h) Change in absorbance

Give a qualitative explanation to the observed change in absorbance of metMb compared to fresh oxy-/deoxyMb?

(i) Type of shift

Is the spectral difference from deoxyMb to metMb a redshift or a blueshift?

(j) Experiment design

You would like to set up an experiment where the absorbance of myoglobin at defined wavelength(s) should be used to measure the level of O₂ binding. Sketch how the absorbance spectra would look like when going from deoxyMb and continuously increasing the concentration of O₂ (draw this for five different O₂ concentrations going from pure deoxyMb to pure oxyMb)

5 Proteins in blood plasma.

```
import pandas as pd
from IPython.display import display
```

In this exercise we will learn how Python is excellent for handling datasets with many data points and how it can be used to apply the same procedure to all the data points at once.

A researcher wants to determine the concentration of two proteins in blood plasma that is suspected to be involved in development of an autoimmune disease. 500 patients and 500 healthy individuals were included in the study and absorbance measurements of the two purified proteins from all blood plasma samples were measured at 280 nm. The molecular weight and extinction coefficients of the two proteins are given in the table below.

Protein	M_w [kDa]	ϵ [M ⁻¹ cm ⁻¹]
1	130	180000
2	57	80000

(a) Load the dataset

Load the dataset as a dataframe from the file protei-blood-plasma.xlsx with pandas. Set data_file_path to the path of the dataset file.

```
data_file_path = ... # Write the path to the dataset file
df = pd.read_excel(data_file_path)
display(df)
```

(b) Calculate concentrations

Calculate the molar concentration of the two proteins in all samples, the light path for every measurement is 0.1 cm.

Always a good idea to assign known values to variables


```
protein_1_ext_coeff = 180000
protein_2_ext_coeff = 80000
l = 0.1
```

💡 Tip

You can set new columns in a DataFrame by just assigning to it

```
df['new_column'] = [1, 2, 3, ..., 42]
```

It can also be set as a computation of a property from another row

```
df['new_column'] = df['current_column'] / 4
```

```
df['protein1_healthy_molar_conc'] = ... # Calculate for concentration in healthy for
protein 1.
... # Your code that updates the data frame with the 3 other new columns.
display(df)
```

(c). Concentrations in mg/mL

Add another set of four columns containing the concentrations in mg/mL.

```
protein_1_mw = 130 * 10**3
protein_2_mw = 57 * 10**3
```

```
df['protein1_healthy_conc'] = ...
df['protein1_patient_conc'] = ...
df['protein2_healthy_conc'] = ...
df['protein2_patient_conc'] = ...

names = ['protein1_healthy_conc', 'protein1_patient_conc', 'protein2_healthy_conc',
'protein2_patient_conc']
display(df[names])
```

(d) Mean concentration

Now that we have the mass concentrations, calculate the mean mass concentration in the four categories.



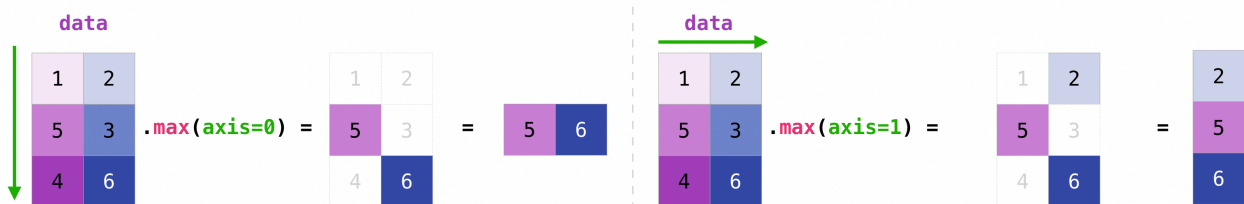
Tip

When displaying the dataframe above we indexed it with names as `df[names]`. We can do the same to compute something over just the four rows.

For example the if we have a DataFrame called `example_df`, we can calculate the mean over the **rows** as:

```
example_df[names].mean(axis=0)
```

Here `axis=0` means that we apply the operation over the first axis which by convention are the rows. The figure below visualizes this



```
mean = ...  
display(mean)
```

(e) Standard deviation

Calculate the standard deviation



Tip

The standard deviation can be calculated using the `.std`-method that works in the same way as the `.mean`-method we used above.

```
std = ...  
display(std)
```

(f) Analyze the results

Consider the following questions

- By comparing healthy individuals with patients, could any of the two proteins would be involved in disease development or be used as a biomarker for diagnosing the disease?
- What additional information does the standard deviation provide in addition to the average value of the protein concentration?

6 Extinction coefficient of human myoglobin

The protein of human myoglobin is given below

```
sequence = ""GLSDGEWQLVLNVWGKVEADIPGHGQEVLIIRLFKGGHPETLEKFDKFKHLKSEDEMKASEDLKKHGA  
TVLTALGGILKKKGHHEAEIKPLAQSHATKHKIPVKYLEFISECIIQVLQSKHPGDFGADAQGAMNKALELFRKDMASNY  
KELGFQG""
```

We want to calculate the *extinction coefficient* of this protein, we have seen that this can be calculated using the formula

$$\epsilon(280\text{nm}) = N_{Trp}\epsilon_{Trp} + N_{Tyr}\epsilon_{Tyr} + N_{Cys}\epsilon_{Cys} \quad (1)$$

Where N_{Trp} is the number of Tryptophan in the protein (and likewise for the other two terms), and the three constants A , B and C are given as

$$\epsilon_{Trp} = 5500 \text{ M}^{-1}\text{cm}^{-1}$$

$$\epsilon_{Tyr} = 1490 \text{ M}^{-1}\text{cm}^{-1}$$

$$\epsilon_{Cys} = 125 \text{ M}^{-1}\text{cm}^{-1}$$

In order to calculate the formula we need to know the count of the relevant residues, we can use Python to get that - for example we can count the number of Tryptophan like so;

```
N_trp = sequence.count("W")
```

(a) Determine the number of residues

In the cell below find the number of residues

```
N_tyr = ... # Your code here  
... # Your code here for the N_cys.
```

You can check what Python has stored each variable by using print

```
print(N_trp)  
print(f"{N_tyr = }") # This is just a way of make a string that looks nice.  
print(f"{N_cys = }")
```

```
2  
N_tyr = 2  
N_cys = 1
```

(b) Calculate the extinction coefficient

Use equation (Equation 1) to calculate the extinction coefficient of human myoglobin.

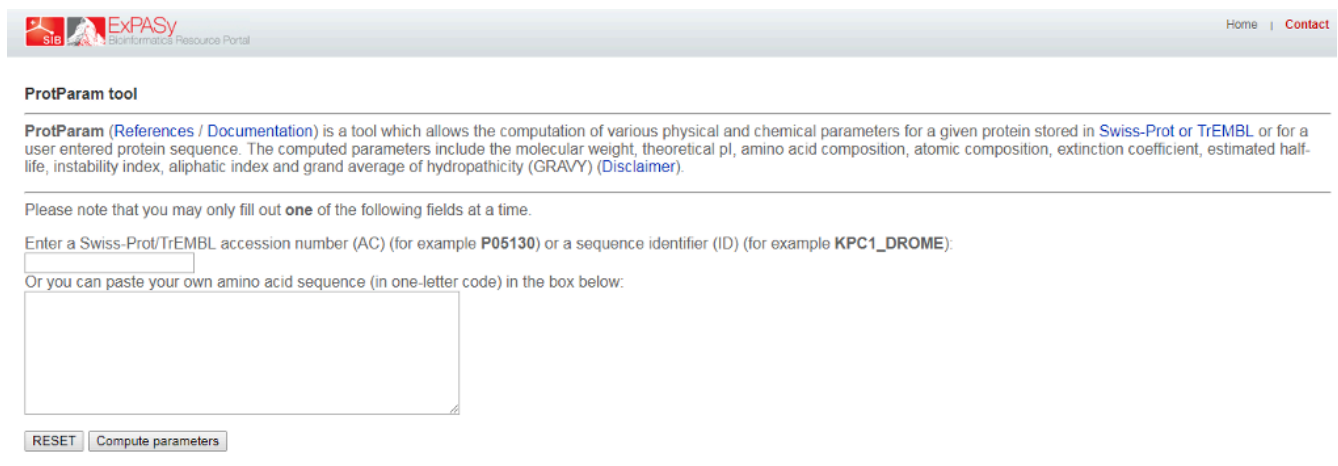
```
eps_trp = 5500
eps_tyr = 1490
eps_cys = 125
```

```
epsilon = ... # Erstat ... med din kode.
print(epsilon)
```

What are the units of this value?

(c) Compare residue count to ProtParam

ProtParam is an online tool that calculates various physical and chemical parameters from a given protein sequence and is used worldwide in research laboratories.



The screenshot shows the ProtParam tool interface on the ExPASy website. The header includes the ExPASy logo and navigation links for Home and Contact. The main heading is "ProtParam tool". Below this, a paragraph describes the tool's purpose: to compute various physical and chemical parameters for a protein sequence. It lists the parameters calculated: molecular weight, theoretical pI, amino acid composition, atomic composition, extinction coefficient, estimated half-life, instability index, aliphatic index, and grand average of hydropathicity (GRAVY). A disclaimer link is provided. A note states that only one field should be filled at a time. There are two input options: a text box for a Swiss-Prot/TrEMBL accession number (AC) or a sequence identifier (ID), and a larger text area for pasting a custom amino acid sequence. At the bottom, there are two buttons: "RESET" and "Compute parameters".

Figure 2: Protpram tool input page.

Go to ProtPram at this link: <https://web.expasy.org/protparam/> and paste the sequence and click **Compute Parameters**. You should then see the calculated parameters, similar to in the image below

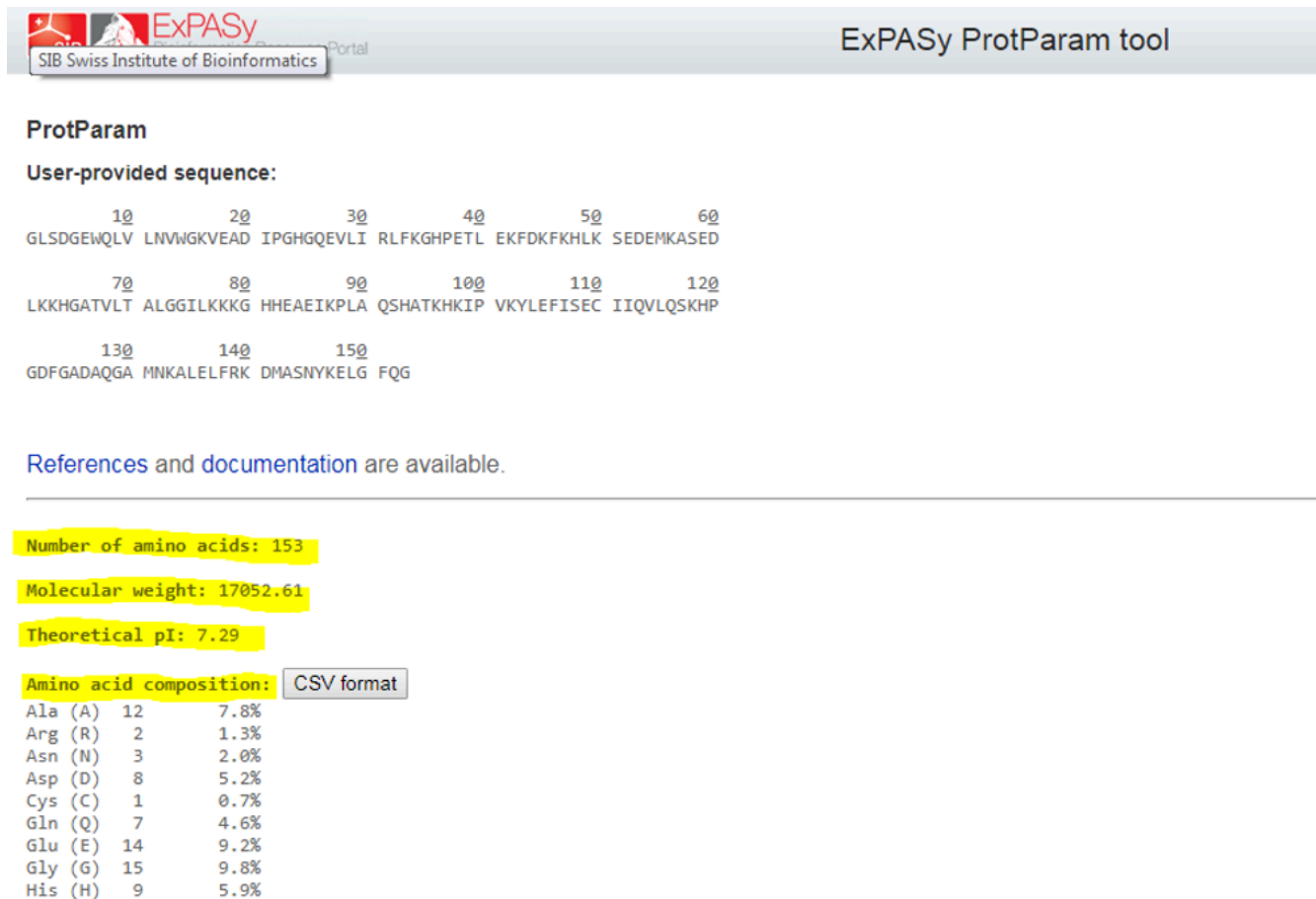


Figure 3: ProtParam tool calculated properties.

On the output page you will see a calculated extinction coefficient. Does that match your calculation? If not, what is the different between the assumptions made?

(d) Calculate the absorbance

Using the extinction coefficient and the molecular weight given by ProtParam, calculate the absorbance at 280 nm of a myoglobin solution at a concentration of 1 mg/mL in a cuvette with a light path of 1 cm.

```
molecular_weight = ... # Find the value on ProtParam (It has units of g/mol)
path_length = ... # Set the value of the path length
concentration = ... # Set the value of the concentration.
```

Remember to convert the concentration to mol/L.

```
A280 = ...
print(A280)
```

This value is what is known as the A280(0.1%) of a protein, i.e. the absorbance of a given protein at a concentration of 0.1% weight/volume (= 1 g/L = 1 mg/mL).

(e) Titin

We have now calculated the extinction coefficient of a protein, now we will make our code more reusable so that it can be applied to other proteins easily.

i Note

A function in Python is a set of instructions, like a recipe, that can be defined and reused multiple times. The syntax is like this

```
def my_function(input_1, input_2):  
    # Some operation  
    some_internal_variable = input_1 + input_2  
    ...  
    return something_weve_calculated
```

Line 1

The def command is used to define the functions name, here my_function, and state its inputs, e.g. the name and ingredients of a recipe.

Line 3

The body of the function defines what it does, the body can have any number of lines of code. Similar to the instructions of a recipe.

Line 5

Finally, the function can return something, like the final product of a recipe.

Note that the function is not executed by doing this, like how a cake isn't baked by writing down the recipe, in order actually use the function it needs to be *called*

```
output = my_function(1, 2)
```

This is also how we have already used other functions like print.

The way of doing so is by defining a function that does the necessary operations for a given sequence. In this way the code can be reused for any sequence.

Finish implementing the body of the function below, note that you have already written all the required code - you just need to copy it into the function.

```
def extinction_coefficient(sequence):  
  
    # Start by counting  
    ...  
  
    # Define the residue extinction coeffiecients  
    eps_trp = 5500  
    eps_tyr = 1490  
    eps_cys = 125  
  
    # Calculate
```

```
epsilon = ...

return epsilon
```

It's always a good idea to check that functions do what we expect, so we can confirm that it gives the same result for human myoglobin as we calculated before

```
epsilon_func = extinction_coefficient(sequence)
print(f"Epsilon: {epsilon}")
print(f"Epsilon from function: {epsilon_func}")
```

The largest known protein is Titin, the cell below loads the sequence of titin and prints a few bits of information about it. (You can also find the full sequence in the dataset `extin-coeff-human-myogl.txt`.)

```
from fysisk_biokemi.datasets import load_dataset
titin_sequence = load_dataset('titin')

print(f'Titin number of residues: {len(titin_sequence)}')
print('First 100 residues')
print(titin_sequence[0:100])
```

Use your function to calculate the extinction coefficient of titin.

```
titin_eps = ... # Your code here.
print(titin_eps)
```

💡 Tip

You don't need to understand the code below, it's just ment to illustrate that knowing some Python will allow you to explore the topics that interest you in more detail.

In general Python is very powerful at letting us explore properties of sequences, for example the cell below calculates number of residues between each Tryptophan in the Titin sequence and plot the distribution.

```
import numpy as np
import matplotlib.pyplot as plt

def get_distance(sequence, letter):
    W_index = np.argwhere(np.array([l for l in sequence]) == letter)
    count = len(W_index)
    distance = (W_index - np.roll(W_index, 1))[1:]
    return distance, count

letters = ['W', 'Y', 'C']

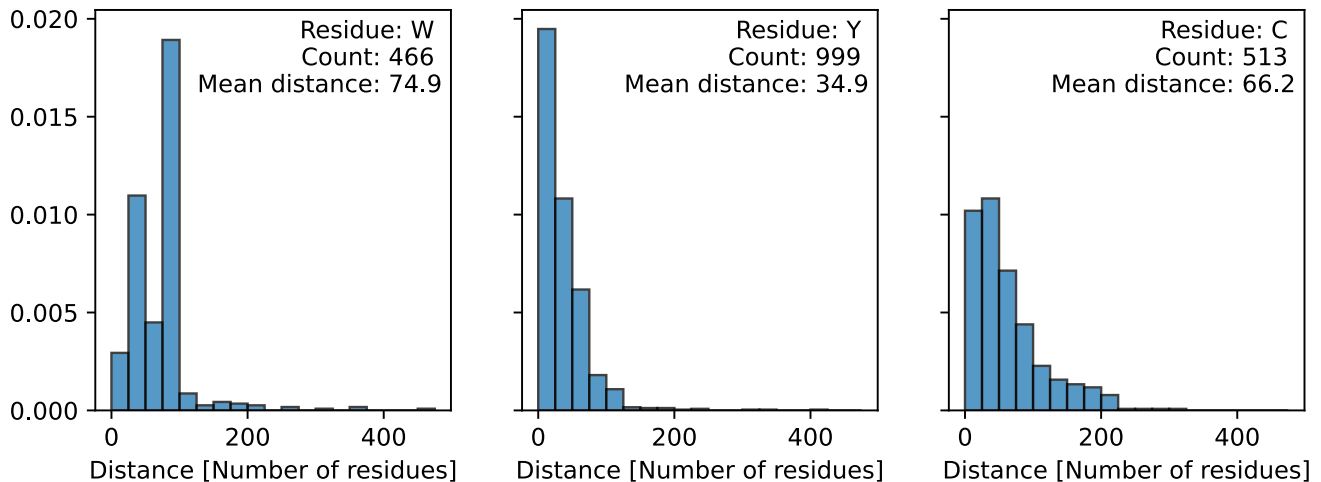
fig, axes = plt.subplots(1, 3, figsize=(3*3, 3), sharey=True)
```

```

axes = axes.flatten()

for ax, letter in zip(axes, letters):
    distance, count = get_distance(titin_sequence, letter)
    ax.hist(distance, bins=np.arange(0, 500, 25),
            edgecolor='black', alpha=0.75, density=True)
    ax.set_xlabel('Distance [Number of residues]')
    info = f'Residue: {letter} \nCount: {count} \nMean distance: {np.mean(distance):.1f}'
    ax.text(0.975, 0.975, info, transform=ax.transAxes, ha='right', va='top')

```



7 Tryptophan absorption and fluorescence

```

from fysisk_biokemi.datasets import load_dataset
from IPython.display import display
import matplotlib.pyplot as plt
import numpy as np
import pandas as pd
pd.set_option('display.max_rows', 6)

```

You investigate a protein that neither contains tyrosine nor cysteine residues. A 37 μM solution gives an absorbance of 0.41 at 280nm at a light path of 1 cm.

(a) Number of residues

How many tryptophan residues does the protein contain?

...

(b) Concentration

You would like to conduct a protein stability experiment at an absorbance of 0,8 (path length 1cm). What concentration should you use?

...

(c) Plot the spectra

The cell below loads extinction and emission spectra for tryptophan in aqueous buffer. (The data files are `trypt-absor-fluor-emission.xlsx` and `trypt-absor-fluor-extinction.xlsx`)

```
df_emission = load_dataset('week47_1_emi')
df_extinction = load_dataset('week47_1_ext')
display(df_emission)
display(df_extinction)
```

	wavelength_(nm)	emission_(AU)
0	280.0	379
1	280.5	396
2	281.0	403
...	...	...
396	478.0	5024
397	478.5	4950
398	479.0	4813

	wavelength_(nm)	molar_extinction_(cm-1/M)
0	219.75	12488
1	220.00	12872
2	220.25	12428
...	...	...
396	318.75	14
397	319.00	18
398	319.25	45

Make plot showing the two spectra

```
fig, axes = plt.subplots(1, 2, figsize=(7, 3), layout='constrained')

# Emission
ax = axes[0]
ax.set_title('Emission')

# Replace ... with your code.
ax.plot(..., ..., label='Emission', color='C0')
```

```

ax.set_ylabel('Emission')
ax.set_xlabel('Wavelength [nm]')

# Extinction
ax = axes[1]
ax.set_title('Extinction')

# Replace ... to plot the extinction spectrum
...

# Customization
ax.set_ylabel('Extinction')
ax.set_xlabel('Wavelength [nm]')
plt.show()

```

(d) Stokes shift

From the plot estimate the Stokes shift

```
estimated_stokes_shift = ...
```

Compare your estimated Stokes shift to the output of the cell below that computes the stokes shift.

```

max_index = np.argmax(df_emission['emission_(AU)'])
emission_wavelength_max = df_emission['wavelength_(nm)'][max_index]

max_index = np.argmax(df_extinction['molar_extinction_(cm-1/M)'])
extinction_wavelength_max = df_extinction['wavelength_(nm)'][max_index]

stokes_shift = emission_wavelength_max - extinction_wavelength_max

print(f"{emission_wavelength_max = :.1f} nm")
print(f"{extinction_wavelength_max = :.1f} nm")
print(f"{stokes_shift = :.1f} nm")
print(f"{estimated_stokes_shift = :.1f} nm")

```

(e) pH and how it changes pHProperties.

Next, you compare the fluorescence emission spectra for your protein at pH 7 and at pH 2. You can find the data in the files `trypt-absor-fluor-ph2.xlsx` and `trypt-absor-fluor-ph7.xlsx` - the cell below also loads these datasets.

The datasets are loaded in a different way than you've seen before - as help to you as this exercise deals with multiple datasets. The datasets could also be loaded with the widget you've seen before

```

df_ph2 = load_dataset('week47_1_ph2')
df_ph7 = load_dataset('week47_1_ph7')
display(df_ph2, df_ph2)

```

	Wavelength(nm)	Fluo_Int
0	311.623037	0.890000
1	314.450262	1.128668
2	315.863874	1.128668
...	...	...
43	367.068063	27.990971
44	368.638743	27.313770
45	370.052356	26.636569

	Wavelength(nm)	Fluo_Int
0	311.623037	0.890000
1	314.450262	1.128668
2	315.863874	1.128668
...	...	...
43	367.068063	27.990971
44	368.638743	27.313770
45	370.052356	26.636569

Make a plot comparing the two emission spectra. Which resembles the spectrum of tryptophan in water most?

```
fig, ax = plt.subplots()

ax.plot(..., ..., label='pH = 2') # Replace to plot at pH = 2
... # Replace to plot at pH = 7

ax.set_xlabel('Wavelength [nm]')
ax.set_ylabel('Flourescence [AU]')
ax.legend()
plt.show()
```

(f) for explain.

Explain why the same protein gives such different spectra at pH 2 and pH 7.

8 The fluorescent protein “mCherry”.

```
import pandas as pd
from IPython.display import display
```

The spectra of many fluorescent proteins can be found at the website: www.fpbases.org. Go to FPbase and search for “mCherry”.

(a) Find parameters.

Find the following parameters for the protein

- Extinction coefficient at absorbance maximum
- Quantum yield
- The organism from which it was originally isolated
- Molecular weight

Save them to separate variables in the cell below.

```
... # Your answers here.
```

(b) Absorbance

What is the absorbance of a 1 μ M solution of mCherry at its absorption maximum at a path length of 1 cm?

```
c = ... # Put the concentration in Molar
l = ... # Path length in cm
A_max = ... # Calculate the adsorbance.
print(f"{A_max = :3.3f}")
```

(c) Extinction coefficient from sequence.

i Note

We will treat the sequence as a str, like any other text, str's are defined like

```
one_line_string = "This is some text, that makes up my string."
```

Which uses "-quotation marks around the text, for longer strings it can be useful to instead use

```
multi_line_string = """This is a very long text, so long in fact that it
takes up multiple lines and I therefore use a slightly different syntax.
To make the string longer I will confess that I am hungry right now.
"""
```

The sequence of the protein is also given. From this determine the extinction coefficient at 280 nm.

Start by taking the sequence from the website and assigning it to the variable sequence in the cell below.

```
sequence = """
Put the sequence in here to make a str with the sequence.
Remember to remove this text as it is not part of the sequence.
"""
```

Now use the sequence to calculate the extinction coefficient, finish the code below (or take the function you implemented in a previous exercise!)

```
# This is "dictionary" with the extinction coefficients of the relevant
# amino acid residues. Dictionaries are indexed with 'keys', so you can retrieve
# the value for W as: ext_residue["W"].
ext_residue = {"W": 5500, "Y": 1490, "C": 125}

# Write code to calculate the extinction coefficient
...
```

(d) Concentration from A280nm

What is the molar concentration of solution of mCherry with an A_{280nm} of 0.45

```
A280 = ...
c_mol = ...
print(c_mol)
```

The excitation and emission spectra can be downloaded as a csv-file by clicking the download icon as highlighted below

Figure 4: Screenshot of the download button on fpbase

(e) Load the dataset

Go to www.fpbase.org and download the spectrum as described above.

Load the dataset as a DataFrame with pandas. Set `data_file_path` to the path of the downloaded dataset file.

```
data_file_path = ... # Write the path to the dataset file
df = pd.read_csv(data_file_path)
display(df)
```

(f) Plot spectra

Make your own plot showing the excitation and emission spectra of “mCherry” using the above data.

```
import matplotlib.pyplot as plt
fig, ax = plt.subplots(figsize=(9, 4))
ax.plot(..., ..., label='Excitation') # Replace ... with your code
... # Replace ... with your code plotting the emission spectrum.
ax.legend()
ax.set_xlabel('Wavelength')
ax.set_ylabel('Extinction & Emission')
plt.show()
```

(g) Stokes shift

Estimate the Stokes shift of mCherry from the plot.

```
estimated_stokes_shift = ... # Your estimation
```

The cell below shows how this could be calculated using Python

```
import numpy as np
# Start by extracting the wavelengths from the DataFrame
wavelengths = df['wavelength']

# Find the indices
ex_max_idx = np.argmax(df['mCherry_ex'])
em_max_idx = np.argmax(df['mCherry_em'])

# Use the index to find the corresponding wavelengths
lambda_ex = wavelengths[ex_max_idx]
lambda_em = wavelengths[em_max_idx]

stokes_shift = lambda_em - lambda_ex

print(f"{lambda_ex = :d}")
print(f"{lambda_em = :d}")
print(f"{stokes_shift = :d}")
print(f"{estimated_stokes_shift = :d}")
```

💡 Tip

If you have two arrays A and B you can find the entry in A corresponding to the largest value in B like this

```
A_at_B_max = A[np.argmax(B)]
```

The `np.argmax` stands for **argument maximum** meaning that it finds the index of the maximum value in a given array. The figure below illustrates this

Figure 5: Illustration of `argmax`

(h) Colors

Compare to a plot of the visual spectrum. What colors are the light that corresponds to the excitation and emission maxima respectively?

Why do you think the protein is called *Cherry*?